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Q1. General principles 1 pts

Which of the following are general principles for writing good R code as discussed in R4Epi?

Correct Answer

- [ ] Comment your code.
- [ ] Scripts should not contain more than 80 lines of code.

Correct Answer

- [ ] Use a style consistently.
- [ ] Write variable names in uppercase.

General answer comments

The coding best practices chapter of R4Epi describes two general principles for writing good R code.

1. **Comment your code**. Whether you intend to share your code with other people or not, make sure to write lots of comments about what you are trying to accomplish in each section of your code and why.
2. **Use a style consistently**. I'm going to suggest several guidelines for styling your R code below, but you may find that you prefer to style your R code in a different way. Whether you adopt my suggested style or not, please find create a style that works for you and your collaborators and use it consistently.

It's true that it's considered a best practice to keep each line of code to 80 characters (including spaces) or less; however, there is no principle that states that scripts should not contain more than 80 lines of code.

Q2. Codebook 1 pts

What do you call a text file that contains detailed information about each variable in your dataset?

- ( ) Encyclopedia
- ( ) Code key
- ( ) Label

Correct Answer

- ( ) Codebook

General answer comments

Even when someone has given the variables informative names, there may still be contextual information about the variables that is important to understand for data management and analysis. Therefore, some data sets will come with something called a **codebook** or **data dictionary**. These are text files that contain information about the data set that are intended to provide you with some of that more detailed information.

Q3. Variable names 1 pts

According to the style guidelines used in this course, which of the following is the most appropriate name for a variable that contains information about the study participants' date of enrollment?

Correct Answer

- date_enrollment
- date.enrollment
- dateEnrollment
- date enrollment

General answer comments

"Variable and function names should use only lowercase letters, numbers, and _. Use underscores (_) (so called snake case) to separate words within a name." **Tidyverse Style Guide** **(https://style.tidyverse.org/syntax.html)** .

Q4. Comments 1 pts

According to the style guidelines used in this course, Which of the following code chunks below contains the most appropriate use of a comment.

Correct Answer

```
df_mdac <- df_mdac %>%
  # Remove rows with missing phone numbers
  filter(!is.na(patient_primary_phone_number))
```

```
df_mdac <- df_mdac %>%
  # Remove rows with missing phone numbers filter(!is.na(patient_primary_phone_number))
```

```
df_mdac <- df_mdac %>%
  # filter
  filter(!is.na(patient_primary_phone_number))
```

```
df_mdac <- df_mdac %>%
  #Remove rows with missing phone numbers
  filter(!is.na(patient_primary_phone_number))
```

General answer comments

Code comments should tell you what the code is trying to accomplish. Further, our style guide tells us that we should put a space between the pound/hash sign and the rest of the comment text. Finally, R will treat any text written *after* the pound/hash sign as a comment — even R code. So, we must either write comments on their own line above the code we are commenting or on the same line as the code we are commenting, but after the code.

Q5. Managing multiple projects 1 pts

You are managing the data for a research project called Care Time. This requires you to import some data from a database, clean that data, and conduct an exploratory data analysis to look for problems with the data (e.g., too many missing responses). According to the style guidelines used in this course, which of the following is the most appropriate file structure to use for this project?

- care_time_data.Rmd
- care_time_analysis.Rmd
- analysis_exploritory.Rmd
- data_clean.Rmd
- data_import.Rmd
- care_time_project.Rmd

Correct Answer

- data_01_import.Rmd
- data_02_clean.Rmd
- analysis_01_exploritory.Rmd

General answer comments

When you are doing data management and analysis for real-world projects you will typically need to break the code up into multiple files. If you don't, the code often becomes really difficult to read and manage. As part of that process, you should separate data cleaning and data analysis into separate files. Data cleaning files should be prefixed with the word "data" and named as `data_[order number]_[purpose]`. Analysis files that do not directly create a table or figure should be prefixed with the word "analysis" and named as `analysis_[order number]_[brief summary of content]`.

Q6. Assignment operator 1 pts

According to the style guidelines used in this course, which of the following code chunks demonstrate the most appropriate way to assign a value to an object?

- `c(1, 2, 3) -> numbers`

Correct Answer

- `numbers <- c(1, 2, 3)`
- `numbers = c(1, 2, 3)`
- `numbers<-c(1,2,3)`

General answer comments

The most appropriate way to assign a value to an object is with the form `[object name] <- [value(s)]`. Best practices dictate that the assignment arrow should have a single space on both sides.

Q7. Spaces 1 pts

According to the style guidelines used in this course, which of the following code chunks demonstrate the most appropriate code for using the mean function to calculate the mean value of the variable **heights**, which is a column in

the data frame called **df**?

Correct Answer

- `mean(df$heights, na.rm = TRUE)`
- `mean( df$heights, na.rm = TRUE )`
- `mean(df$heights,na.rm=TRUE)`
- `df.mean(heights, na.rm = TRUE)`

General answer comments

The **tidyverse style guide** **(https://style.tidyverse.org/syntax.html)** dictates that we should always put a single space after a comma and an equal sign, but we should not put spaces inside or outside a regular function call.

Q8. Pipes 1 pts

According to the style guidelines used in this course, which of the following code chunks demonstrate the most appropriate code for creating the **result** object?

Note: I know that many of you won't yet know what the code below does. That's ok. To answer this question correctly, you need only focus on code style and best practices.

- `result <- select(filter(data, incident_num > 1), cancer)`

  `more_than_one_incident <- filter(data, incident_num > 1)`

- `result <- select(more_than_one_incident, cancer)`

Correct Answer

```
result <- data %>%
  filter(incident_num > 1) %>%
  select(cancer)
```

General answer comments

According to the style guidelines used in this course, we *generally* want to use the pipe operator instead of nesting functions or performing a series of steps that created intermediate objects that aren't of interest.

Q9. One function per line 1 pts

According to the style guidelines used in this course, which of the following code chunks demonstrate the most appropriate code for creating the **result** object?

Note: I know that many of you won't yet know what the code below does. That's ok. To answer this question correctly, you need only focus on code style and best practices.

Correct Answer

```
result <- data %>%
  filter(incident_num > 1) %>%
  select(cancer)
```

```
result <- data %>% filter(incident_num > 1) %>%
  select(cancer)
```

```
result <- data %>% filter(incident_num > 1) %>% select(cancer)
```

General answer comments

According to the style guidelines used in this course, we generally want to use the pipe operator in such a way that each line of my code does one, and only one, thing.

Q10. R markdown files 1 pts

In general, what file type will we use for writing and storing R code the majority of the time?

- R scripts

Correct Answer

- R markdown files
- R project files
- Plain text files

General answer comments

In the chapter on **R Scripts** (https://www.r4epi.com/r-scripts.html#r-scripts), you learned how to create R scripts – plain text files that contain R code and comments. These R scripts are kind of a big deal because they give us a simple and effective tool for saving, modifying, and sharing our R code. If it weren't for the existence of R markdown files, we would probably do all of the coding in this book using R scripts. However, R markdown files *do* exist and they are AWESOME! So, we're actually going to suggest that you use them instead of R scripts the majority of the time.

Bonus 0 pts

Which of the following lines of code would be most appropriate for creating a level one header in your R markdown file?

Correct Answer

- `# Data cleaning`
- `## Data cleaning`
- `#Data cleaning`
- `<h1> Data cleaning`

General answer comments

The R markdown **cheat sheet** (https://www.rstudio.com/resources/cheatsheets/) referred to in the **chapter on R markdown** (https://www.r4epi.com/r-markdown.html) shows you how to use the pound/hash sign to create headers in R markdown documents.

You can turn text that is typed *inside* of a code chunk in an R markdown document into a *comment* by prefixing it with one or more pound/hash signs and a space.

You can turn text that is typed *outside* of a code chunk in an R markdown document into a *header* by prefixing it with one or more pound/hash signs and a space. For a level one header, you would prefix it with one pound/hash sign and a space. For a level two header, you would prefix it with two pound/hash signs and a space. And so on.

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